

# Davi Barros de Rezende

## SENIOR SOFTWARE ENGINEER · GENERATIVE AI AND MISSION-CRITICAL SYSTEMS

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Timezone: UTC-3, one to two hours ahead of US Eastern, so a full working-day overlap with US teams and roughly four hours with CET. Available as contractor or through an EOR.

### SUMMARY

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Software engineer with about 5 years of experience and 2,900 authored commits across nine production systems. Tech Lead and Owner of OOInfo (ooinfo.org), a collaborative data platform with 495 endpoints and 76 tables, where I built the MCP server that exposes the database to AI agents entirely on my own. I bring to AI in production the measurement discipline of someone who also shipped, alone from first commit to last, an on-premise SCADA system delivered to an industrial client. I treat PostgreSQL as an architectural tool: multi-tenant isolation through Row-Level Security, queues inside the database, and full-text search over a million records, instead of adding infrastructure.

### SKILLS

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**Languages:** TypeScript, JavaScript, Python 3, SQL, PL/pgSQL, C++ (ESP32), Bash

**Generative AI and LLM:** LLMs in production (OpenAI, Anthropic, DeepSeek, OpenRouter, Gemini), Model Context Protocol (MCP), OAuth 2.1, AI agents, tool calling, multi-agent systems, multiprovider LLM gateway with fallback cascade, SSE streaming, structured output with JSON Schema, prompt engineering, LLM as a Judge, evals, hallucination mitigation, per-tenant cost control, ONNX Runtime, scikit-learn

**Backend and APIs:** Node.js, NestJS, Fastify, Express, Next.js (App Router, Server Actions), Django REST, FastAPI, REST, OpenAPI, JSON-RPC, idempotent webhooks, modular monolith, BullMQ, PostgreSQL-backed job queues, Server-Sent Events, WebSocket, Socket.IO, MQTT

**Frontend and mobile:** React 18 and 19, React Native, Expo, Vite, TanStack Query, Zustand, TailwindCSS, shadcn/ui, React Hook Form, Zod, Recharts, i18n, accessibility (ARIA), Tauri

**Data:** PostgreSQL 16, Prisma, TypeORM, MongoDB, SQLite, Redis, InfluxDB, Row-Level Security, PL/pgSQL functions and triggers, versioned migrations, full-text search (tsvector, pg\_trgm, unaccent, GIN), keyset pagination, FOR UPDATE SKIP LOCKED, query optimisation

**Observability:** OpenTelemetry (traces and metrics), OTel Collector, Prometheus, Grafana, Tempo, Loki, structured logs correlated by trace\_id, label cardinality control, liveness/readiness health checks, allowlist-based PII redaction

**Infrastructure and quality:** Docker, Docker Compose, GitHub Actions, GHCR, Coolify, Nginx, Caddy, Linux, VPS, on-premise and air-gapped deployment, Jest, Vitest, Cypress, Playwright, pytest, integration tests against a real database in CI, chaos testing, soak testing, ADRs

**Security:** OAuth 2.0 and 2.1, Authorization Code with PKCE, OpenID Connect, Keycloak SSO, JWT, RBAC, row-level ACL, multi-tenant isolation enforced in the database, bcrypt, SHA-256, AES-256-GCM, rate limiting, IDOR testing, audit trails, LGPD

**Payments:** Stripe webhook signature verification written from scratch (HMAC-SHA256, timing-safe comparison, replay window), Mercado Pago, Pix, charge idempotency

**Mission-critical and embedded:** SCADA, Siemens S7-1500 PLC, S7Comm protocol, binary protocol reverse engineering, fail-safe design, watchdogs, GNSS-RTK, NTRIP, RTCM3, u-blox ZED-F9P, sensor fusion, Extended Kalman Filter, Bayesian optimisation, geospatial (GDAL, KML)

### EXPERIENCE

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#### OOInfo · Software Engineer, Tech Lead

09/2025 – Present · Contract · Remote · Live at ooinfo.org, today a team of 4 developers

- Technical lead of OOInfo (ooinfo.org, live in production): 1,537 of the repository's 2,865 commits across 18 authors, including the founding commit, with the Tech Lead and Owner role recorded in CONTRIBUTING.md. NestJS, React, PostgreSQL 16, Prisma, Redis and BullMQ, totalling 495 REST endpoints, 76 tables, 131 migrations and 13 asynchronous queues.
- Designed and built single-handedly the MCP server that exposes the platform to external AI agents (33 of the module's 37 commits): 15 JSON-RPC tools, OAuth 2.1 with Dynamic Client Registration, tokens stored only as SHA-256, and 4 scopes that can only narrow the existing access-control layer.
- Built the generative AI layer as its dominant author (132 of 226 commits): multiprovider gateway with a three-step fallback cascade under a global time budget, SSE streaming with an incremental parser, an agentic tool loop, and audited quota.
- Rewrote the read path for lists holding 1,010,025 records, which suffered 6.6 s per page and a 60 s timeout on text search, replacing EAV scanning with materialised columns (weighted tsvector, GIN pg\_trgm, unaccent), keyset pagination and versioned Redis caching.
- Set up three-environment CI/CD on GitHub Actions and Coolify, with 39 Cypress suites, 31 of them blocking in CI.

## **LABIC (Federal University of Goiás) · Software Engineer**

01/2026 – Present · Contract · Goiás · Remote · Part-time · Public selection process (RTVE call 33/2025), 2-year delivery schedule

- On the citizen self-service portal for the state water utility of Goiás, in a team of five to six, I own the service-request, metering-report and privacy modules: 27 of the repository's 179 commits. Keycloak SSO and the Next-to-Vite migration belong to other people.
- On the lowcodejs core, the open-source framework behind those portals, I have 110 of 2,694 commits, almost all in React Native and TypeScript.

## **Brametal · Software Engineer, Industrial SCADA system**

12/2025 – Present · Contract · Remote and on-site · Delivered through my own company

- Built alone, across 368 of 368 commits over nine months, the SCADA system that drives 52 variable-frequency drives and reads 56 load cells during destructive testing of 80-metre transmission towers. React, Node.js, WebSocket, 100 endpoints and 26 tables over PostgreSQL, Redis and InfluxDB, speaking S7Comm directly to a Siemens S7-1500 PLC, shipped as a signed Windows executable.
- Owned architecture, on-site commissioning and operator training for the test rig, not just the software.
- Reverse-engineered the PLC's 15 KB memory contract byte by byte from third-party sources, proving each structure by offset CRC rather than name similarity. A test that parses the control engineer's own export caught an address error that would have killed the safety heartbeat during a live test.
- Designed the fail-safe boundary between the supervisory system and the PLC: a heartbeat in an isolated process flipping one bit every 100 ms, so the PLC cuts traction on its own if the backend dies. Added a longevity lab with seven chaos scenarios that fails the build and distinguishes product failure from measurement failure.

## **Orbya · Senior Software Engineer**

04/2026 – 07/2026 · Contract · Remote · B2B Customer Intelligence and Revenue Orchestration platform

- Architected as principal author, with 398 of 428 commits, a multi-tenant B2B SaaS on Next.js and PostgreSQL 16: 36 models, 34 migrations, 108 Server Actions and 37 screens in three and a half months.
- Isolated tenant data at the database layer with Row-Level Security across 18 tables, 6 PL/pgSQL functions and an application role without BYPASSRLS, so forgetting the scope returns zero rows instead of leaking data. Proved the isolation with an automated cross-tenant IDOR test running in CI against real PostgreSQL.
- Built a deterministic anti-hallucination gate that matches every number the model writes against the context it received, as a precondition for publishing the insight, alongside an LLM-as-a-Judge eval bench that elects the production model on quality per dollar.
- Removed Redis and WebSocket from the stack using pg\_notify, LISTEN and Server-Sent Events, with the analysis queue inside PostgreSQL itself. 1,026 Vitest blocks against a real database in CI, and pull-based deployment via GHCR.

## **EMFLORA · Automation Analyst (full time)**

11/2024 – 07/2026 · Espírito Santo, Brazil · Hybrid · Mid-level 08/2025, Junior 01/2025, Intern 11/2024

- Built alone, across 364 commits and 41,000 lines of Python, the precision navigation firmware of an agricultural post-hole digger on a Raspberry Pi: GNSS and IMU fusion with a 15-state Extended Kalman Filter, custom UBX, NMEA and RTCM3 parsers, and an NTRIP client with failover.
- Eliminated a magnetic heading fault that had blocked operations for two months by building a seven-script calibration lab and deriving the parameters from measurement, reaching a final mean error of 3.1°.
- Raised the navigation score from 64.7 to 88.6 on the hardest scenario through Bayesian optimisation of the filter parameters over a simulator fed by real field datasets.
- Designed the RTK infrastructure behind the rover and the fleet app: an ESP32 base station with a u-blox ZED-F9P publishing RTCM3 over NTRIP, a Fastify backend converting KML plans into waypoints, and an offline-first React Native app.

## **FAPES (Espírito Santo State Research Foundation) · Junior Web Developer**

01/2023 – 02/2025 · Remote · Part time

- Built and maintained the foundation's web applications in PHP and Laravel.

## **Independent · Web Developer (contract)**

05/2021 – 01/2023 · Remote · Direct clients

- Web development for direct clients, from requirements gathering through deployment.

## **SELECTED PROJECTS**

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**LowCodeJS** (lowcodejs.org, open source): designed the row-level authorisation plugin contract (RowAccessGuard), with a three-valued decision composed across independent guards and ACL filter pushdown into the MongoDB query. Delivered by a pull request merged upstream; the interface is still on main today.

**Signfy**: an AI calligraphic-signature SaaS built alone (413 of 418 commits), with 81 Fastify endpoints and 455 test cases. Developed a curvature-spectrum descriptor for handwritten strokes using orthonormal DCT-II, then publicly retracted a result of my own after an adversarial audit showed it was measuring the derivative operator rather than the signature.

**Observability Demo** ([github.com/Daviqr1/observability-demo](https://github.com/Daviqr1/observability-demo), public): a NestJS service instrumented end to end. OpenTelemetry traces crossing three processes (API, external HTTP service and queue worker) landing in Grafana as a single 25-span trace, with context surviving the Redis queue. Prometheus metrics with deliberately bounded cardinality (route template, never the raw URL; unmatched requests collapse into `unmatched`), logs correlated by `trace\_id` in Loki, and allowlist-based PII redaction in two independent layers. The whole stack comes up with `docker compose up`.

**Bank churn classification** (public on GitHub): RandomForestClassifier over a bank customer base, with a published article. I document alongside it the label-leakage limitation found in later review.

## EDUCATION AND LANGUAGES

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**Systems Analysis and Development**, Anhanguera, in progress, expected 2027

**Computer Science**, Descomplica, in progress, expected 2029

**Mechanical Engineering**, IFES (Federal Institute of Espírito Santo), attended, not completed

**Languages:** Portuguese native. English with advanced technical reading and writing, backed by the roughly 55,000 lines of technical documentation I wrote.